Glucose Excursion Twin: architecture

Offline training (top) is separate from the live twin engine and dashboard (bottom). Real data never ships to the public app; demo patients are synthetic or a small CC BY-NC-SA slice.

Offline: data and training

Open data sources
CGMacros (PhysioNet)
45 adults, Dexcom G6
CC BY-NC-SA 4.0
ShanghaiT2DM (Figshare)
100 T2D, Libre H
external test only

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Data engineering
de-interpolate 1-min release
to real 5-min readings
range / spike / gap checks
regular grid, gaps kept
as missing (never filled)
provenance per table

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Leakage-safe dataset
features use data ≤ t only
labels: any reading ≥180
in (t, t+120 min]
≥80% future coverage
scored only if G(t) < 180
unit tests for leakage

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Models
Kalman twin: identify p, τ,
noise by multi-step error
Hybrid: GB residual + event
Baselines: persistence,
ridge, gradient boosting
conformal + Platt on held-out

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Evaluation
5-fold CV by participant
bootstrap CIs, paired tests
ablations: inputs, personal-
isation, missing, noise,
intervals, alert workload
external: ShanghaiT2DM

trained artefacts:
twin_params.json, hybrid.joblib, serving.json

Online: digital twin engine and clinician dashboard

Virtual patient
static record (EHR-like):
age, sex, HbA1c, group,
diagnoses, medications
CGM stream, replayed
5 min at a time
synthetic patients S-01..06
labelled SYNTHETIC

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Input validation
duplicate timestamps
implausible values
missing reading
stale data (>15 min)
insufficient history
end of stream
poor quality → no risk
shown, explicit warning

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Twin state engine
hidden state per patient:
glucose, inflow, baseline
Kalman update each 5 min
forecast 0–120 min
Monte-Carlo P(any ≥180)
hybrid residual correction
conformal 90% interval
one engine per patient

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Alert logic
alert if risk ≥ threshold
threshold chosen on
calibration patients
(≤1 false alert/day)
60-min refractory period
outcome tracked: excursion
followed / false alert

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Clinician dashboard
review queue by risk
forecast fan + 180 line
data-quality flags
twin states explained
alerts and outcomes
what-if (model scenario)
workload calculator

Research prototype: not a medical device, not clinically validated, not validated on Indian patients. Forecasts are scenarios for discussion, not treatment instructions.